

EXAM #2 INFORMATION

Where and when: Wednesday November 12, 5:45-7:15 PM, Birge 145.
(Notice that the location is different than Exam 1).

Structure of exam: There could be true/false questions and there could be some short answer questions (e.g. what is the $\liminf/\limsup$ /subsequential limit set of a given sequence). The rest of the exam will be proof questions similar to those on the homework sets and the activities.

How to prepare: Redoing the activities and homework problems is a good start. It is also a good idea to read the posted solutions for the homework problems. Some practice problems are given below. After that, the textbook has additional problems in each section that you could use as practice (selected hints and answers are given in the back).

Material tested: The exam will cover Chapter 2 in Ross and the formal definition of the limit.

The important Concepts & Learning Objectives for Exam 2 are given below.

Chapter 2:

- Know the formal definition for a sequence to *diverge* to $\pm\infty$ and know how to write a proof showing a given sequence diverges to $\pm\infty$.
- State the definition of a *monotonic sequence*, in particular, what it means for a sequence to be *increasing* or *decreasing* (**Definition 10.1**).
- Determine whether or not a given sequence is monotonic (**Example 10.1**).
- State properties of monotonic sequences related to convergence (**Theorem 10.2, Theorem 10.4**).
- Understand what a *subsequence* is (**Definition 11.1**).
- Understand conditions under which subsequences converge or diverge and find those subsequences (**Theorem 11.2, Examples 11.3-11.4**).
- State the relationships between sequences and subsequences, including the **Bolzano-Weierstrass Theorem (Theorem 11.3-11.5)**.
- Understand the definitions of $\limsup$ and $\liminf$ for a sequence (**Definition 10.6**).
- Understand how the $\limsup$ and $\liminf$ of a sequence relate to the limit (**Theorem 10.7**).

- Understand the definition of a *subsequential limit* and properties of the set of subsequential limits (**Definition 11.6, Theorem 11.8**).
- For a given sequence, find its set of subsequential limits and determine its $\limsup$ and $\liminf$ (**Examples 11.5-11.12**).
- State the formal definition of a *Cauchy sequence* (**Definition 10.8**).
- Understand the relationship between convergent sequences and Cauchy sequences (**Theorems 10.9-10.11**).

Chapter 3:

- Know the ϵ/δ definition of $\lim_{x \rightarrow a} f(x) = L$.
- Given a concrete function f and concrete numbers a , L , and $\epsilon > 0$, be able to find a $\delta > 0$ that makes the statement “If $0 < |x - a| < \delta$, then $|f(x) - L| < \epsilon$ ” true.
- Be able to write a ϵ/δ limit proof for linear functions.
- Know the sequence version of $\lim_{x \rightarrow a} f(x) = L$.

Sample Proof Questions

- (1) Prove: If $\lim s_n = 0$ and $s_n > 0$ for all n , then $\lim \frac{1}{s_n} = +\infty$.
- (2) Prove: If $\lim s_n = s$ and (s_{n_k}) is a subsequence, then $\lim s_{n_k} = s$.
- (3) Prove: If (s_n) is a bounded sequence and (s_{n_k}) is a subsequence of (s_n) , then $\liminf s_n \leq \liminf s_{n_k}$.
- (4) Prove: If (s_n) converges, then (s_n) is Cauchy.
- (5) Prove, directly from the definition of a Cauchy sequence, the following statement: If (s_n) is a Cauchy sequence and there exists a subsequence (s_{n_k}) such that $\lim s_{n_k} = s$, then $\lim s_n = s$.
- (6) Prove using the ϵ/δ definition: $\lim_{x \rightarrow 3} 7x = 21$.